Confidence Interval (CI)

A **confidence interval** is a range of values that is likely to contain the true population parameter

Example population mean.

Instead of estimating a population parameter with a single number, a confidence interval provides a **range of plausible values**.

Simple Definition

"Based on my sample data, what range of values is likely to contain the true population mean?"

Suppose you want to estimate the average height of all students in a college.

You take a sample and find:

Sample mean = 70 inches

Sample size = 15

95% Confidence Interval = (66.96 73.04)

Interpretation:

We are 95% confident that the true average height of all students lies between 66.92 **inches and** 73.07 **inches.**

Student	Hieght
St-1	61.0
St-2	63.0
St-3	64.0
St-4	66.0
St-5	68.0
St-6	69.0
St-7	70.0
St-8	70.0
St-9	71.0
St-10	72.0
St-11	74.0
St-12	76.0
St-13	77.0
St-14	78.0
St-15	83.1

$$x = \bar{x} \pm ME$$

$\bar{x} = Sample\ Mean = 70$

ME $= Margin\ of\ Error = 3.07$

$$x = 70 \pm 3.07$$

$$x = [66.92, 73.07]$$

$$ME = z^* \times SE$$

$z^* = Critical\ value (Depends\ on\ Confidence\ Level) 95\%$

$SE = Standard\ Error$

$Number\ of\ Sample\ n\ = 15$

$$standard\ Error (SE) = \frac{\sigma}{\sqrt{n}}$$

$\sigma = Standard\ Deviation$

$$Standard\ Deviation\ (\sigma) = \frac{\sum \sqrt{(61-70)^2} + (63-70)^2 + (64-70)^2 + (66-70)^2 + (68-70)^2 + (69-70)^2 + \cdots \ldots \ldots + (83.1-59)^2}{15-1}$$

$Standard\ Deviation = \boxed{6.09}$

$$standard\ Error = \frac{\sigma}{\sqrt{n}}$$

$$standard\ Error = \frac{6.09}{\sqrt{15}}$$

$$standard\ Error = 1.57$$

Confidence Level	(z^*) Critical Value
80%	1.282
85%	1.440
90%	1.645
95%	1.960
98%	2.326
99%	2.576
99.90%	3.291

$$ME = z^* \times SE$$

$$ME = Critical\ value(95\%) \times 1.57$$

$$ME = 1.960 \times 1.57$$

$$ME = 3.07$$

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$\bar{x} = Sample\ Mean$

$\text{ME} = Margin\ of\ Error$

$$x = 70 \pm 3.07$$

$$x = [66.92, 73.07]$$

Confidence Level vs Width

Confidence Level	Interval Width
90%	Narrower
95%	Wider
99%	Widest

Higher confidence means:

- More certainty
- Wider interval

Lower confidence means:

- Less certainty
- Narrower interval

Confidence Level (CL)	Critical Value ($z*$)	($ME = z* \times SE$)	Calculated CI [$\bar{x} \pm ME$]
80%	1.282	1.282 x 1.572 ME = 2.02	[68.79, 72.83]
85%	1.44	1.440 x 1.572 ME = 2.26	[68.55, 73.07]
90%	1.645	1.645 x 1.572 ME = 2.59	[68.22, 73.40]
95%	1.96	1.960 x 1.572 ME = 3.07	[66.92, 73.07]
98%	2.326	2.326 x 1.572 ME = 3.66	[67.15, 74.47]
99%	2.576	2.576 x 1.572 ME = 4.05	[66.76, 74.86]
99.90%	3.291	3.291 x 1.572 ME = 5.17	[65.64, 75.98]

Why Do We Need Confidence Intervals?

A sample provides only an estimate of the population value.

For example:

Population: 1,000,000 people

Sample: 1,000 people

Sample mean income: ₹40,000

Another sample might give ₹39,500 or ₹40,500.

A confidence interval shows the range within which the true population mean is likely to fall.

Structure of a Confidence Interval

A confidence interval consists of:

Point Estimate±Margin of Error

Example:

40,000±1,500

Therefore,

(38,500, 41,500)

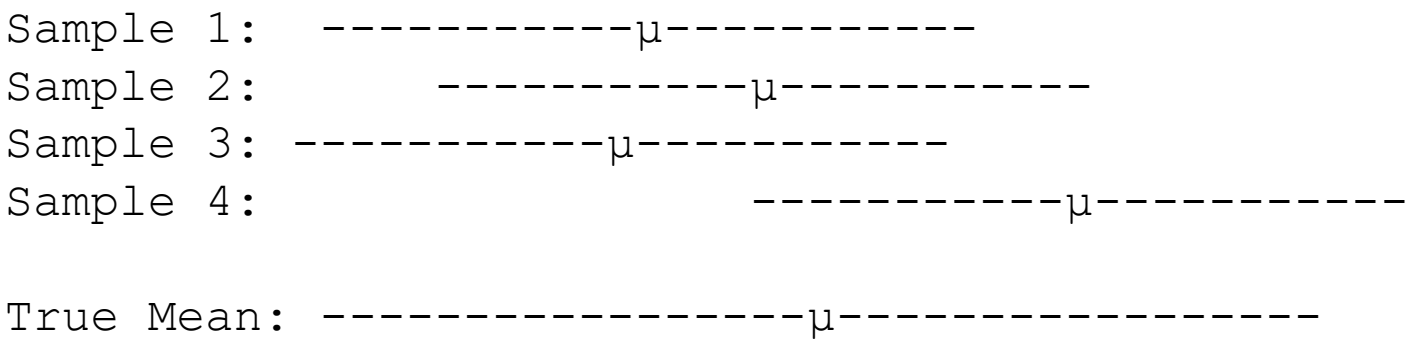
Point estimate: ₹40,000

Margin of error: ₹1,500

Confidence interval: ₹38,500 to ₹41,500

What Does "95% Confidence" Mean?

If we repeatedly take samples and construct a 95% confidence interval each time, about 95% of those intervals will contain the true population mean.



Factors Affecting Confidence Interval Width

1. Sample Size (n)

Larger sample size → Narrower interval → More precise estimate

Example:

- n = 50 → wider interval
- n = 1000 → narrower interval

2.small the confident interval narrower is the interval precise estimation

3. Data Variability

More variability in the data leads to a wider interval.

Example Calculation

A sample of 100 students has:

Mean score = 70

Standard deviation = 10

Sample size n=100n = 100n=100

Calculate the CI

- For a 90% confidence interval:
- For a 95% confidence interval:
- For a 95% confidence interval:

$$ci = \bar{x} \mp z \left(\frac{\sigma}{\sqrt{n}}\right)$$

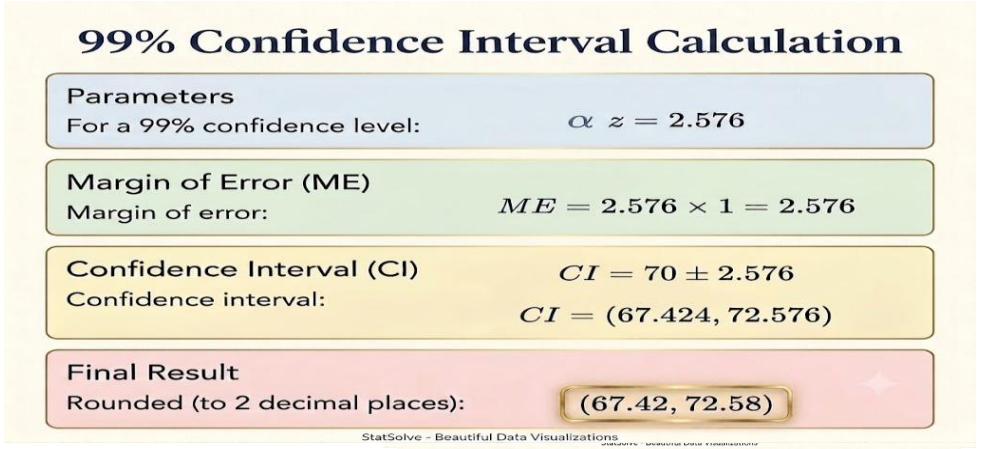
First calculate the **standard error**:

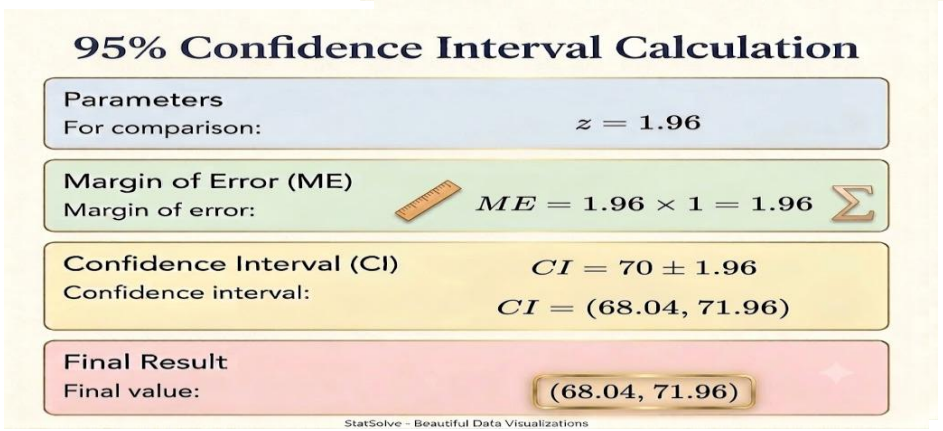
$$SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{100}} = \frac{10}{10} = 1$$

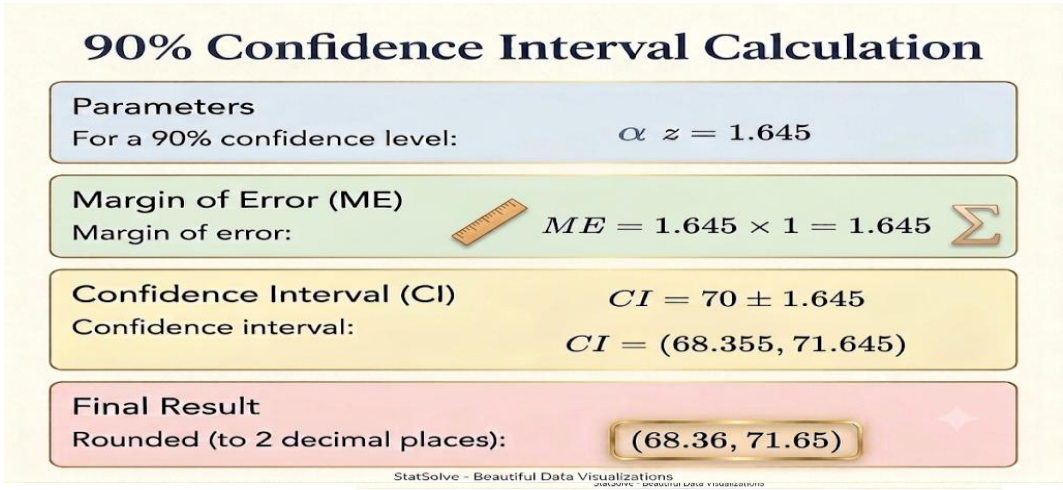
$$ci = \bar{x} \mp z\left(\frac{\sigma}{\sqrt{n}}\right)$$

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Relationship with Hypothesis Testing

Suppose a 95% confidence interval is:

$$(68.04,\ 71.96)$$

Testing:

$$H_0 : \mu = 75$$

Since 75 is **outside** the interval:

✅ Reject H_0 at the 5% significance level.

If the claimed value is inside the interval:

✅ Fail to reject H_0.